SecureDT - Installation Guide

Intelligent and Secure Task Scheduling for Digital Twin-Enabled Edge-Cloud Networks

1. System Prerequisites

• Python 3.10 or newer (64-bit)
• pip (bundled with Python)
• A web browser (Chrome / Edge / Firefox)

2. Install Dependencies

From the project root:

```
python -m pip install -r requirements.txt
```

Key packages: Flask, Flask-Cors, pandas, numpy (>=2.0), scikit-learn, reportlab, fpdf2.

3. Generate Datasets & Train the AI Model

```
python scripts/generate_dataset.py --tasks 10000
python scripts/train_ai.py
```

Six synthetic datasets are written to datasets/ and the Random Forest model to models/risk_model.pkl.

4. Start the Application

```
python app.py
```

Open http://127.0.0.1:8000 in your browser.

5. Alternative One-Command Launchers

Windows: run_windows.bat
Linux/macOS: ./run.sh (chmod +x first)

6. Troubleshooting

	Resolution
319) on read_csv	pip install --only-binary=:all: --force-reinstall --no-deps numpy==2.
	Change config.PORT or set PORT in .env
	Run scripts/train_ai.py; the engine falls back to rules otherwise